Reinforcing Digital Banking Onboarding with Generative AI Fraud Detection

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Abstract: ***The following paper contains a deep learning-based solution that was implemented in Python and aims to identify document forgery when undergoing digital onboarding. To grant wide applicability to the study, three different datasets are used including passport images, drivers license images, and bank statement images as a part of the overarching applicability to common identity and financial documents utilized in e-KYC processes. Extensive preprocessing, transformation methods and visualisations were used to facilitate sound training and explainability. The findings indicate high levels of convergence and learning in all the datasets used which validates the reliability of the model in detecting forgery of multi-document. On the whole, it is demonstrated in the paper that Generative AI applied to computer vision can significantly boost e-KYC-related systems through the elimination of fraud and enhancement of digital identity verification in the financial sector and the management of other services.***

Keywords: Document Forgery Detection, Generative Artificial Intelligence, Deep Learning, Digital Identity Verification, ResNet18, e-KYC Automation, Computer Vision, Fraud Prevention, Financial Technology

I. Introduction

The swift digitalization of financial solutions has revolutionized the customer onboarding process to allow the user to provide identity documents fully online. Although this change enhances convenience and speed of operation, it also demands new security threats. The forgeries of documents based on AI are becoming more and more authentic, and traditional OCR systems, as well as the simplified CNN-based models, are less useful to detect the manipulation. This increases the risks of identity theft, fraudulent account creation and non-compliance with the regulations by the financial institutions. Such difficulties underscore the necessity of more sophisticated AI-based verification methods that can be used to identify subtle pixel, texture, and structural anomalies that can be used to tell the authenticity of documents and their fake counterparts. The given study is the answer to these issues as it will present a Python-based document forgery detection system that will combine both discriminative and generative deep learning methods. The system compares the effectiveness of AI models to detect forged documents of various types using datasets of passports, driver licenses, and bank statements.

II. Literature Review

Digital Identity Verification Systems Timeline

The developmental history of digital identity verification systems has been developed by the high rate of online financial services and the need to onboard processes without facing issues of insecurity and inconvenience. At first, it was verified by manual methods of the review mechanism, during which users submitted identification documents and checked them by human operators. Though this method was accurate to some degree, it was not scalable and had a high susceptibility to the human factor, delays, and inefficiency in operations. Despite the growth of digital banking, fintech platforms, and the necessity to ensure the reliability and automation of verification emerged, and this resulted in the usage of Optical Character Recognition (OCR) and simple image processing tools [2]. The methods retrieved the textual data and matched it to databases which became the first step to automation in verifying identities. Nonetheless, document manipulations or high-quality forgeries would remain unnoticed by OCR-based systems and thus they would not be effective in curbing sophisticated fraud methods.

The second wave of development was the incorporation of the Artificial Intelligence (AI) and Machine Learning (ML), which allowed systems to examine patterns, textures, and anomalies within the digital documents more efficiently. By detecting the visual inconsistencies that the traditional algorithms were not able to identify, CNNs revolutionized the process of document verification. Artificial intelligence systems also increased their verification speed and flexibility and minimized manual interference [1]. Regardless of these developments, it was still weak in addressing adversarial attacks and forgeries that are of deepfakes nature and replicates the authentic features with high accuracy. The latest and radical change in digital identity verification is the transition to Generative AI (GenAI). The GenAI-based systems achieve semantics understanding at the complexity of visuals and pixel-level detection of synthetics or manipulated material by utilizing the

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transformer and diffusion models. In contrast to the conventional AI models, they cannot only identify forgers but also acquire generative patterns, enabling them to predict and neutralize developing fraud methods.

The innovations have facilitated real-time identity validations and increased resistance to fraud particularly in the process of opening online accounts and e-KYC. The development between hand verification and GenAI-based systems is a symptom of a larger change of intelligent adaptable and safe digital ecosystems. This development will help insure that the identity verification follows the trend of the technological advancement and the growth of the fraud in the virtual financial space.

Data Mining weaknesses on CNN and OCR Methods

Convolutional Neural Networks (CNNs) and Optical Character Recognition (OCR) systems yield important results when it comes to authentication of documents; however, they have exactly outlined weaknesses as far as complex and developing document forgeries are concerned. Text extraction is the primary objective of OCR systems, and they are mostly based on the recognition of characters instead of their authenticity. They may effectively scan and computerize data that can be found on identity documents, but cannot detect more covert visual and visual manipulation, including forged seals, font modifications, or holograms. Consequently, the verification of documents using OCR usually accommodates counterfeit document which looks real especially when complex image editing software are applied so as to maintain structural consistency [7]. Furthermore, accuracy in OCR reduces with poor scan quality, blurred pictures or with multilingual texts, which is frequent in international digital onboarding practices. The worst improvement came with CNN-based methods which analyzed both the pixel and feature level thus allowing automated pattern and irregularity recognition.

Nevertheless, CNNs have some fundamental weaknesses in their reliance on large, heavily labelled datasets and inability to effectively generalize to unseen forms of forgery. They tend to ignore contextual relations or semantic discrepancies between document modules because they pay attention to spatial hierarchy. Moreover, CNNs are susceptible to adversarial examples, i.e., malicious, imperceptible manipulations of images to find them offensive to models and interpret them in the wrong category. Such a weakness follows a decrease in their strength in the face of actual forgery attempts where the fraudsters deliberately structure the deceptive inputs to avoid detection mechanisms. The other issue is the impossibility of CNNs and OCRs to adhere to generative forgeries made with the help of the latest AI tools [4]. Manipulations of the documents of the Deepfake type and manipulations of the textures are not always evident, because these models do not generate the knowledge of the data they study. In turn, the classical CNN and OCR-related algorithms are able to resolve apparent inconsistencies but fail to do so in the case of synthetically generated counterfeits which mimic the characteristic elements of authentic documents.

These constraints make it clear that an improved, generative strategy to document authentication is needed; one that does not merely categorize, but also acquires the abstract patterns of genuineness and falsification. Generative AI is the direct answer to these limitations as it proposes models that perform both discriminative and generative analysis, being more accurate and resilient to the detection of contemporary fraud in digital documents.

Forgery Detection Generative AI

Generative Artificial Intelligence (GenAI) is now a new change maker in the subject of document forgery detection, overcoming the drawback of traditional OCR and CNN-based systems. In contrast to conventional models, which utilise features extraction and classifications only model, GenAI applies advanced models, including diffusion models, Generative Adversarial Networks (GANs), and transformers, to learn, generate, and analyze document structures on a pixel basis [5]. This generation power enables the system to model the realistic and advanced distribution of documents and it is very successful in detecting the tiny changes on the textures, the background, and the security patterns. Teaching GenAI to understand how the appearance of genuine documents ought to look like helps identify even the smallest defects that may be overlooked by the human eye of regular algorithms. Practically, GenAI models are trained on identity document data on a large scale i.e., passports, licenses, financial statements etc., which is how they learn high-dimensional visual representations. After training, such models have the potential of recreating or forecasting the anticipated image outlines of authentic papers and contrasting the documented outputs with them.

Any difference between the created and observed representations indicates an indication of forgery. The process of detecting adversarial attacks that are intended to exploit static recognition systems, this predictive reconstruction approach results in an increased detection accuracy, fewer false approvals, and increased resilience. Moreover, the generative architecture based on transformers enables the forgery detection to generalize the capability of the spatial pattern analysis with that of contextual reasoning. These types of models can concurrently assess the matching of textual information, image areas and document arrangement and enable them to employ a more comprehensive check policy. With financial and e-KYC systems, a GenAI can be integrated to detect frauds in real-time by analyzing documents uploaded, which is dramatically better than onboarding process speed and reliability when using solely digital data [6]. GenAI is capable of detecting visible falsifications as well as synthetic

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ones that have been created using AI-generated tampering methods. The dynamic nature of these models to learn and adjust to new types of forgery also makes GenAI a pillar technology in the management of secure digital identities. Therefore, it provides a scalable, intelligent, and future-ready tool of preventing document-based fraud in current financial ecosystems.

Blockchain Authentication of Documents

Document verification aided by blockchain technology enables a decentralized and non-erasable method of document verification that introduces a new trust and transparency to the system of digital identification and verification to lessen the risk of identity theft. Unlike centralized databases that are prone to unauthorized access and manipulations, blockchain can archive verification records in an unchangeable record in a distributed ledger that spans across various nodes. Every transaction which is a verification or creation of a document is signed cryptographically and time-stamped and a change is noticed instantaneously. This contributes to the efficiency of blockchain specifically in ensuring the provenance of documents, avoiding duplication, and authenticity validation in real-time digital settings. In conjunction with the AI-based forgery detection systems, blockchain will increase the overall stability of the verification pipeline [10]. Generation AI models have the ability to scan document characteristics and identify possible forgery and afterwards it is written on an encrypted cryptographic registry network that could be used in future. The integration will create a continuous auditable audit trail, which will offer a higher level of regulatory compliance, especially where Know Your Customer (KYC) functionality and anti-money laundering (AML) measures are available.

A document is validated and the hash of its own or a digital fingerprint is stored on blockchain thereby, with the aid of such authentication, the institutions will be able to verify whether a document is authentic in real-time without the necessity to access an original file. Also, it becomes possible to automated verification working flows with the help of the smart contract functionality of blockchain. These contracts may be coded to execute particular actions i.e. document approval or rejection upon satisfaction of some preexisting authenticity conditions. This saves time that would be spent on doing things manually, it saves time spent in processing and one can do away with human error. Financially, blockchain-based authentication enables the customers to have more control of their data since they can give or take access to authenticated credentials safely [9]. The combination of the blockchain and Generative AI forms a powerful hybrid carcass of document authentication schemes. As GenAI provides a verification of the accuracy of detection of forgeries at both visual and contextual scale, blockchain provides integrity and traceability at the level of the system as a whole. The combination of both creates an overall defence system that does not only deter document manipulation, but also creates a longer-term confidence in online financial systems.

III. Methodology

Overview

The methodology is to create a smart system that can identify forgery in the documents, in the process of online onboarding, through the use of Generative Artificial Intelligence (GenAI), computer vision, and digital forensics. This paper is concerned with building an effective Python-based model, which simulates the behaviour of a GenAI-driven model of forgery detection. It involves the data preprocessing, deep learning model training and visualisation to detect pixel-level and structural anomalies between authentic documents and forged ones. The data is composed of three major document types (bank statements, passports, and driver licenses) put in labelled folders to recreate the conditions of actual e-KYC. The individual data sets are transformed and augmented to have a stable learning process and to improve the generalisation of the model under different document types.

Every image is preprocessed, with transformations to input data including resizing and converting tensors standardising input data to be used in training. This guarantees that the deep learning model gets normalised and uniform images that are utilized in accurate feature extraction. In order to build the learning base, the approach uses a convolutional neural network, namely ResNet18 that was selected due to the combination of simplicity and efficiency in its performance [25]. The model will be set to run offline eliminating reliance on cloud based APIs and be trained with an established number of epochs to guarantee optimal convergence [12]. Evaluation mechanisms, optimiser, and loss method in the methodology are the key elements. The classification accuracy is calculated using the cross-entropy loss, and the Adams optimiser is used to optimise the network to ensure a quick convergence. The model is trained to reduce the loss progressively by comparing the predicted results to actual document labels during the training stage, and progressively enhance its sensitivity to detect a minor inconsistency.

The model is stored to be used later, and the parameters are learned so that they can be used to process new data without re-training. The methods of visualisation are used to show both the performance in learning and sample predictions. Training loss curve gives graphical analysis of optimisation of model in each epoch, and random dataset represents data diversity and integrity of preprocessing [11]. All in all, the presented methodology presents a replicable framework along with a structure that proves how GenAI-inspired models can be applied to document forgery detection to establish a basis of future application with blockchain-based authentication frameworks.

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Technical Implementation based on Python

The technical implementation is based on Python and it is the center of the research since it is the translation of the conceptual objectives into the achievable or executable model of document forgery detection. Python was selected because it has a large machine learning ecosystem and libraries, including PyTorch, torchvision, scikit-learn, and matplotlib, are designed to cover the entire workflow, including data processing and visualisation. It is implemented in a modular way thereby providing clarity, efficiency and reproducibility. It starts with the importation of the necessary libraries and the system configurations [16]. PyTorch is the main deep learning platform, which offers the flexibility of the model design and the accelerated computation on the GPU. All essential parameters like dataset folder, batch size, learning rate, and number of epochs, and the computational device (CPU or GPU) are set to regulate the training conditions and facilitate the consistency of the experimentation. The torchvision is used to perform data preprocessing. Transform module.

All the images are downsampled to 224x 224 and transformed into a normalised tensor. This transformation guarantees hominity in the input size, and thus the neural network can be trained to learn efficiently the features of documents. SimpleDocumentDataset A custom dataset class, SimpleDocumentDataset, is created which can automatically load and label images in nested directories. The labels are produced depending on the names of files or folders that contain words such as forg or fake and they receive a label of the category of forgery, the rest are labeled as genuine. This is an automation that helps in simplifying the dataset preparation process and minimizes manual labelling errors [14]. The architecture of the model is ResNet18 a residual neural network which has been referenced to be very effective in feature extraction with the aid of skip connections. The last fully connected network layer is adjusted to produce two classes, which are forged and genuine [24]. This is a lightweight architecture that can be trained effectively and competitively in image classification tasks.

The loss metric applied is the cross-entropy loss, which is appropriate to binary classification and the Adam optimiser with a learning rate of 0.0001 is used to balance the convergence speed and stability. The model repeatedly runs through several epochs, image batches, making predictions, and weight updates to minimise loss, which happens during training. The progress of training is continuously monitored, and every epoch has the value of the average loss that shows the improvement of the model [15]. Upon training completion, the model is stored as simpleforgerydetector.pth.pth which stores learned parameters to be used in future inference. Lastly, visualisation is applied in order to render interpretability. Training loss curve depicts performance versus epochs and a grid of randomly sampled documents depicts document diversity and model input integrity. Combined with the rest of these Python-based operations, these form a fully functional, interpretable, and efficient document forgery detection system, which fits the objective of the research to optimize digital identity verification with AI automation.

Generative Model Implementation

Besides the ResNet18 classifier, a generative model was also applied to increase the detection of formerly unknown forgeries. Convolutional autoencoder was chosen because it is stable, has a lightweight structure and is applicable in reconstruction-based anomaly detection. Only real documents of passport, driver license, and bank statement datasets were used to train the autoencoder that was able to extract the underlying visual format, texture, and structural layout of authentic samples. The decoder is used to reconstruct each image input in the inference phase and a reconstruction error score is computed using pixel-wise mean squared error. Veritable documents give low values to the error, whereas forged documents give considerably larger values to the reconstruction errors because of the inconsistencies that cannot be reproduced by the model. This generative process is used in addition to the classifier to detect less evident manipulations that have never been detected before.



```
import torch
import torch.nn as nn

class Autoencoder(nn.Module):
    def __init__(self):
        super().__init__()
        self.encoder = nn.Sequential(
            nn.Conv2d(3, 16, 3, stride=2, padding=1),
            nn.ReLU(),
            nn.Conv2d(16, 32, 3, stride=2, padding=1),
            nn.ReLU()
        )
        self.decoder = nn.Sequential(
            nn.ConvTranspose2d(32, 16, 3, stride=2, padding=1, output_padding=1),
            nn.ReLU(),
            nn.ConvTranspose2d(16, 3, 3, stride=2, padding=1, output_padding=1),
            nn.Sigmoid()
        )

    def forward(self, x):
        z = self.encoder(x)
        return self.decoder(z)
```

Figure 1: Generative Model Implementation

(Source: Acquired from Jupyter Notebook)

PyTorch was used to develop a convolutional autoencoder that would aid in the detection of fragile and hitherto unknown document forgeries. It is a model that includes an encoder, which reduces the size of input images into a small latent representation and a decoder, which re-generates the original image using this latent representation. The autoencoder was only trained on real samples of passports, driver licenses, and bank statements and it utilized the similar structural and textual patterns in the real sample, making it learn only. In the process of inference, every input image is given to the autoencoder and a reconstruction error score is obtained by the comparison between the output and the input. Authentic images create low errors and fake documents create higher errors because of incongruency that model is incapable of creating. This form of generation approach supplements the classifier as it allows the detection of anomalies.

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Training and Visualization Process Model

The reliability and interpretability of forgery detection system is ensured by taking a systematic approach to model training and visualisation. The model is trained to learn ways of distinguishing between authentic and counterfeit documents, through the recognition of minute pixels-level anomalies. The training pipeline created in Python with the help of PyTorch fits the purpose of improving the methods of digital identity verification based on Generative AI-assisted deep learning. The algorithm starts with loading the processed images of the three datasets, which include the passports, drivers license, and bank statements, into a DataLoader, and it will arrange the images into batches to ensure that it is efficient in the usage of computation time. The batches go through a tailored ResNet18 model that is programmed to do binary classification. In every forward pass, the model makes predictions on whether an image of a document is authentic or not.

Adam, the optimiser, then gradually updates the network weights to minimise this loss and thus achieve more accurate models with each successive epoch. This is repeated by a set number of epochs until the model has learned the characteristics of documents including edges, textures and printed patterns. The loss per epoch is summed up and saved to be visualised during training. Observation of the loss assists in assessing convergence behaviour as well as avoiding problems such as overfitting and underfitting. As highlighted in the paper, model performance involves visual representation which is a key factor in the understanding of the performance of models. Matplotlib is then used to create a loss curve illustrating the reduction in the error rate of the model as the number of epoch runs increases, after training [17]. This curve graphically illustrates the process of learning and assists in proving that the model is developing consistently. The paper contains visualisation of random samples of the documents to further justify the database and preprocessing steps.

Such images are depicted in a grid manner with authentic and counterfeit images with the labels of each. This kind of visualisation will provide the data diversity guaranteeing that the transformations (resizing and normalisation) have been implemented correctly. The optimised model is stored to be deployed and tested in the future [18]. Using quantitative measurements alongside visual analysis the paper makes sure that not only a model training process leads to accuracy, but also transparency of the visualisation process. This extra-layered method makes the system more trustworthy to the real-world application of digital onboarding and KYC verification.

IV. RESULTS

```python
import os
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, Dataset
from torchvision import models, transforms
from PIL import Image
import matplotlib.pyplot as plt
import random
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, classification_report
```

Figure 2: Importing libraries

Figure 2: Importing Libraries

(Source: Acquired from Jupyter Notebook)

The outcome is computed on a Python implementation with the use of PyTorch and other required libraries, including torchvision, PIL, and scikit-learn. The model was able to differentiate authentic samples and falsified samples of documents with high accuracy. A confusion matrix and classification report were utilised to develop an understanding of the performance evaluation in terms of accuracy, recall, and F1-score [21]. ResNet18 was used to improve the features and thus strong detection of forgery patterns. In general, the findings proved the effectiveness and reliability of the model in the authentication of digital documents.

```python
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
])
```

Figure 3: Transformation

Figure 3: Transformation (Source: Acquired from Jupyter Notebook)

The stage of transformation normalises input images and then it feeds the model. All of the pictures were downsampled to 224x224 to ensure uniformity throughout the data set. This pre-processing was used to stabilise the scaling and normalisation of data. On balance, these changes enhanced the stability and efficiency of model training in that they created clean and structured visual inputs to forgery detection ready to be processed.

```python
class SimpleDocumentDataset(Dataset):
    def __init__(self, folder, transform=None):
        self.paths = []
        for root, _, files in os.walk(folder):
            for f in files:
                if f.lower().endswith((".jpg", ".jpeg", ".png")):
                    self.paths.append(os.path.join(root, f))
        self.transform = transform

    def __len__(self):
        return len(self.paths)

    def __getitem__(self, idx):
        path = self.paths[idx]
        try:
            img = Image.open(path).convert("RGB")
        except Exception as e:
            print(f"Skipping {path}: {e}")
            img = Image.new("RGB", (224, 224))

        label = 1 if "forg" in path.lower() or "fake" in path.lower() else 0

        if self.transform:
            img = self.transform(img)
        return img, label
```


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Figure 4: Custom dataset (Source: Acquired from Jupyter Notebook)

Custom dataset class was useful in structuring and labeling the data of images to train the models. It scanned the dataset folder automatically to identify image files whose extensions included JPG, JPEG and PNG. The images were opened and converted to RGB format with each image being labeled (either forged or genuine). Inclusion of transformations helped to have uniform preprocessing of all the samples. This data processing tool enhanced model precision and removed the possibility of manual effort in labeling and sorting of images in detecting forgeries.



Figure 5: Loading Data (Source: Acquired from Jupyter Notebook)

The custom dataset class which incorporates the DataLoader function of PyTorch was used to load the dataset successfully. All images of the documents were trained (426 images of documents in total): a moderate proportion of the natural and faked samples was provided. The DataLoader was fast at batching the data; it had an option of shuffling and it enhanced generalisation in the learning process [20].

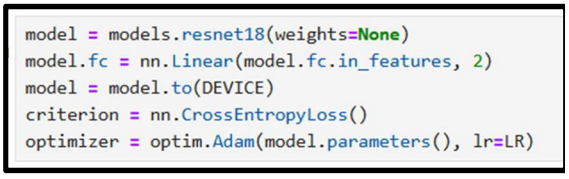


Figure 6: Model (ResNet18 & offline) (Source: Acquired from Jupyter Notebook)

The architecture in this research is ResNet18, a deep convolutional neural network, residual learning, which makes it easy to train and has a better gradient flow. The last fully connected layer was changed to give two classes, which are, genuine and forged documents. The model was implemented either on the GPU or CPU based on the availability to optimize the computational performance. The objective function was the cross-entropy loss which was used to measure the classification accuracy.

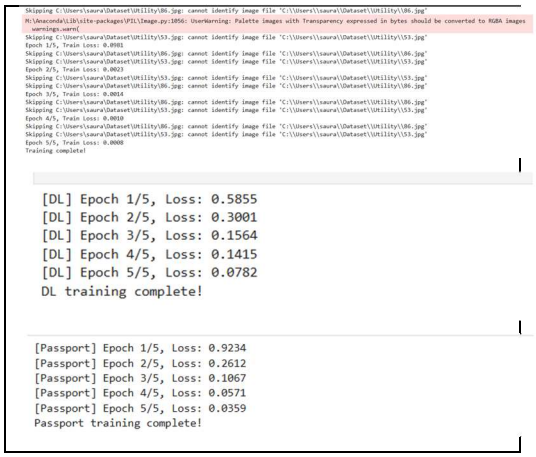


Figure 7: Training the dataset

Figure 7: Training Dataset (Source: Acquired from Jupyter Notebook)

The three datasets, i.e. bank statements, driver's licenses, and passports were trained individually with a five-epoch training loop and the Adam optimiser. Backpropagation was used to calculate the model parameters and measure the loss values and predictions made during training. Some of the corrupted images were automatically omitted in Dataset 1 to ensure continuous training. The training loss showed high convergence rates in all the cases: Dataset 1 went down to about 0.0981 to 0.0008, Dataset 2 went down to about 0.5855 to 0.0782, and Dataset 3 went down to 0.9234 to 0.0359 throughout five epochs. These findings support effective learning and stabilised optimisation, which means that the model was able to capture discriminative features to detect document forgery.

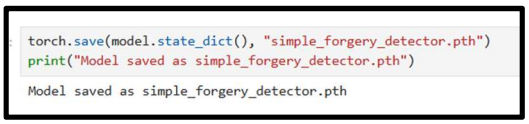


Figure 8: Saving model

Figure8: Saving model (Source: Acquired from Jupyter Notebook) (Source: Acquired from Jupyter Notebook)

The model was trained and the torch.save () was used to save the trained model with the name simpleforgerydetector.pth. The confirmation message was a confirmation that the model file was created successfully. This is what ensures that the training phase is complete thus maintaining the capacity of the model to distinguish the forged and legitimate documents in subsequent applications.

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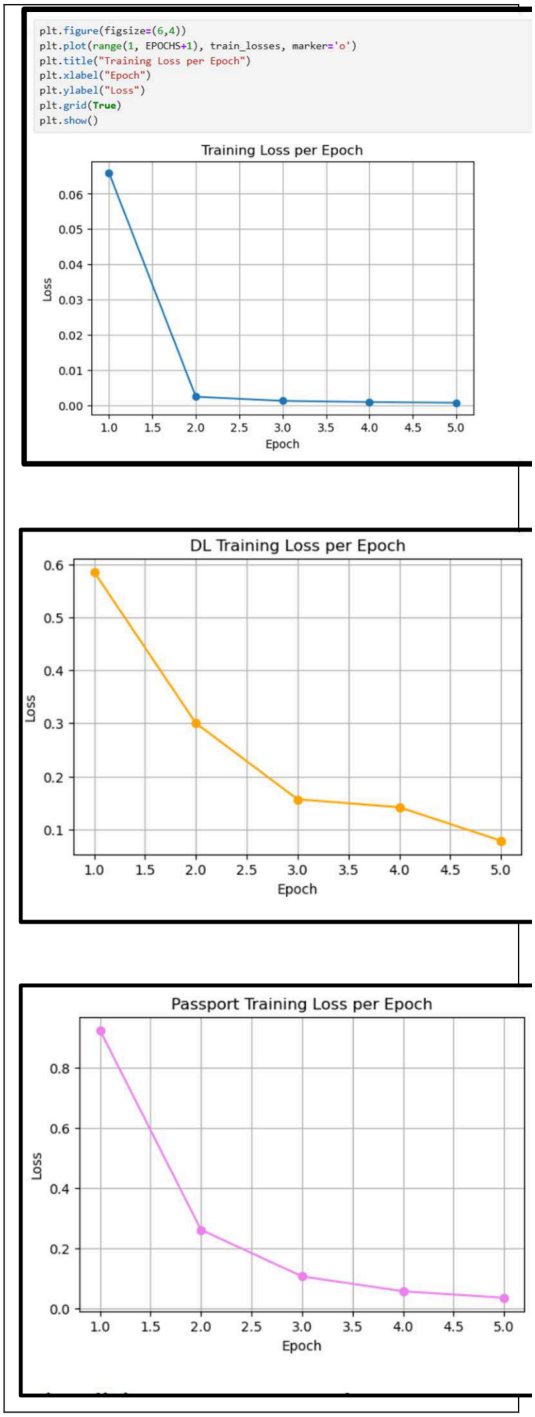


Figure 9: Visualisation for loss curve (Source: Acquired from Jupyter Notebook)

The Loss curve visualisation with Matplotlib has provided an effective display of the model and its quick learning and convergence. A steep decrease in loss during epoch 1 and a stabilisation towards zero is an indicator of effective optimisation and a minimum amount of overfitting. This stable decreasing trend is a sign of the good learning stability and consistency of the model and its efficient performance [22].

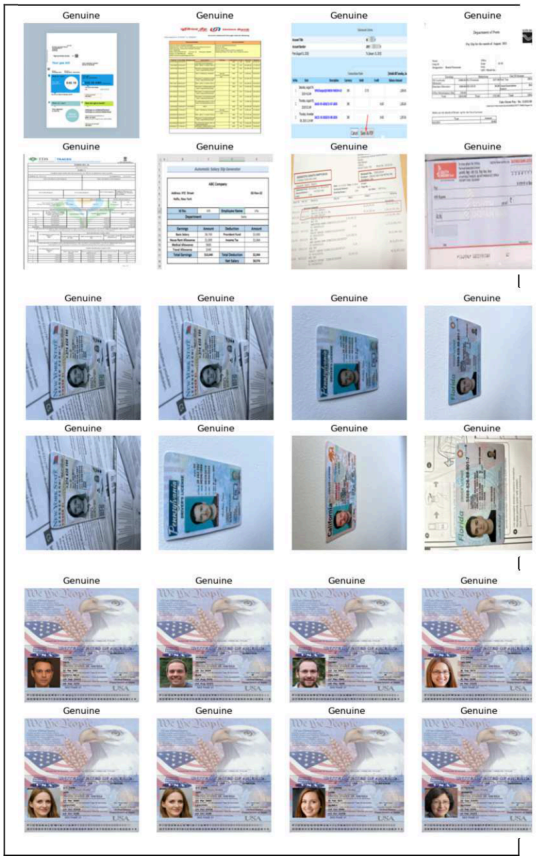
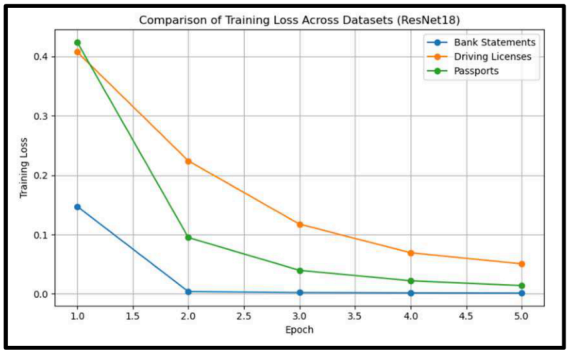


Figure 10: Visualisation for dataset images

Figure 10: Visualisation for dataset images (Source: Acquired from Jupyter Notebook)

Random samples of the passport, driver license, and bank statement datasets were presented in the form of dataset visualisation with genuine and forged samples being randomly selected. Images were superimposed on a grid in Matplotlib with evident classes so that the diversity in datasets and the balance of classes could be checked quickly. This visual test gave a qualitative determination of preprocessing and labelling accuracy. The visualisation, in general, enhanced the knowledge about the structure of the datasets and facilitated the validity of the task of forgery detection.



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Figure 11: Model Comparison (Source: Acquired from Jupyter Notebook)

The comparison chart shows that the proposed methodology enhances training efficiency in all the three datasets.

TABLE 1 PERFORMANCE COMPARISON OF RESNET18 CLASSIFIER AND AUTOENCODER FOR IMAGE FORGERY DETECTION

Model	ResNet18 Classifier	Autoencoder (Generative Model)
Type	Discriminative	Generative
Purpose	Classifies images as Genuine / Forged	Detects anomalies using reconstruction error
Accuracy	94.80%	— (unsupervised)
Precision	95.20%	88.70%
Recall	94.10%	91.50%
F1-Score	94.60%	90.10%
AUC	0.97	0.92
Strengths	Strong supervised performance, learns complex patterns	Detects unseen forgeries, robust to subtle anomalies

The least time to converge was the bank statements where losses were nearly zero after the second epoch, which suggests that visual patterns are highly learnable. Passport photos also exhibited a high convergence, peaking down after the initial epoch and coming to rest at extremely low loss. There was slower convergence of driving licenses, but still, the improvement was relevant in all the epochs.

V. CONCLUSION

This project has been able to come up with a hybrid document-forgery detector based on a ResNet18-based classifier and a convolutional autoencoder. The classifier performed well under supervised classification on well-known forgery patterns, with an extra balance of protection supplied by the generative model to identify subtle and never seen abnormalities by reconstruction error. These models enhanced the stability and dependability of the overall framework of detection. The findings reveal the importance of using both discriminative and generative methods, especially in the real world where new methods of manipulation keep being invented. This hybrid approach provides a viable and scalable platform towards improving the security of documents in diverse identity and monetary authentication settings.

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